

CHM 120

Boyle's Law

Final Report

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Lesson	Boyle's Law
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Course	CHM 120
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Test Your Knowledge

Classify each statement as true or false.

True	False
1 Force is equal to mass times acceleration. Pressure is the ratio of force per unit area. A graph of volume versus $1/\text{pressure}$ displays a linear relationship.	2 Gas particle collisions decrease as volume decreases. Boyle's law describes the relationship of a gas's pressure and temperature at constant volume.

Match each term with the best description.

Constant value used in Boyle's law	1	k
International System unit of force	2	Newton
Area of a circle	3	$A = \pi r^2$
International System unit of pressure	4	Pascal

Exploration

When volume of gas in a closed system decreases, the pressure of the gas in the system ____.

- ☐ decreases
- ☒ increases
- ☐ remains constant

When pressure is plotted against volume, the relationship is ____.

- ☐ linear
- ☐ nonlinear
- ☒ exponential
- ☐ static

Pressure is dependent upon ____.

- ☐ mass
- ☐ acceleration due to gravity
- ☐ surface area of the force
- ☒ All of above

A Newton is the International System unit of pressure.

☐ True

☒ False

Exercise 1

State Boyle's law, including the associated equation(s).

B / U

25 Word(s)

Boyle's Law (Equation) : $PV = k$; $P_1V_1 = P_2V_2$

Assuming constant mass and temperature, this law states that as the volume of a gas decreases, its pressure increases.

What are sources of error for this experiment?

B / U

35 Word(s)

Some sources of error would be, not starting the syringe at the same spot each time, and having the books wiggle causing the syringe to possibly lower more than it would have without the wiggle

Would you expect your experimental results would change if a different gas, such as CO₂, were used instead of air? Explain your answer.

B / U

37 Word(s)

There may be an effect on how much the books lower the syringe. Using CO₂ we could possibly see that the syringe would lower more, or maybe it takes more to lower being CO₂ instead of air

How does Graph 2 illustrate Boyle's law?

B / U 19 Word(s)

Graph 2 shows that pressure is inversely proportional to volume. As the pressure is increased, the volume is decreased.

A gas sample in a Boyle's law apparatus is compressed to a volume of 22.0 mL. If its original volume was 37.0 mL at 12,000 Pa, what is the new pressure of the gas sample. Show your calculations.

B / U 50 / 10000 Word Limit

Using $P_1V_1=P_2V_2$

We rearrange to solve for P_2
 $P_2 = 2.02 \times 10^4 \text{ Pa}$ --> this makes sense because we know from Boyle's Law that as the volume decreases, as it does in this problem, the pressure increases, which it also does in this problem. The pressure goes from 12,000 to 20,000 Pa, essentially.

Data Table 1: Calculating Force

Book #	Mass of Book (kg)	Force (N)
1	1.17kg	11.5N
2	0.930kg	9.12N
3	0.735kg	7.21N
4	0.581kg	5.70N
5	0.848kg	8.32N

Data Table 2: Calculating Surface Area

Diameter of Syringe (mm)	30.mm
Diameter of Syringe (m)	0.030m
Radius of Syringe (m)	0.015
Surface Area of Syring Tip (m ²)	0.00071

Data Table 3: Calculating Pressure

Book #	Pressure (Pa)
1	16197.Pa
2	12845.Pa
3	10155.Pa
4	8028.2Pa
5	11718.Pa
1 and 2	29042.Pa
1,2, and 3	39197.Pa
1,2,3, and 4	47225.Pa
1,2,3,4, and 5	58943.Pa

Data Table 4: Volume and Inverse Volume

Book #	Volume: Trial 1 (mL)
1	50.mL
1,2	45.mL
1,2,3	41.mL
1,2,3,4	38.mL
1,2,3,4,5	35.mL

Book #	Volume: Trial 2 (mL)
1	54.mL
1,2	48.mL
1,2,3	45.mL
1,2,3,4	40.mL
1,2,3,4,5	37.mL

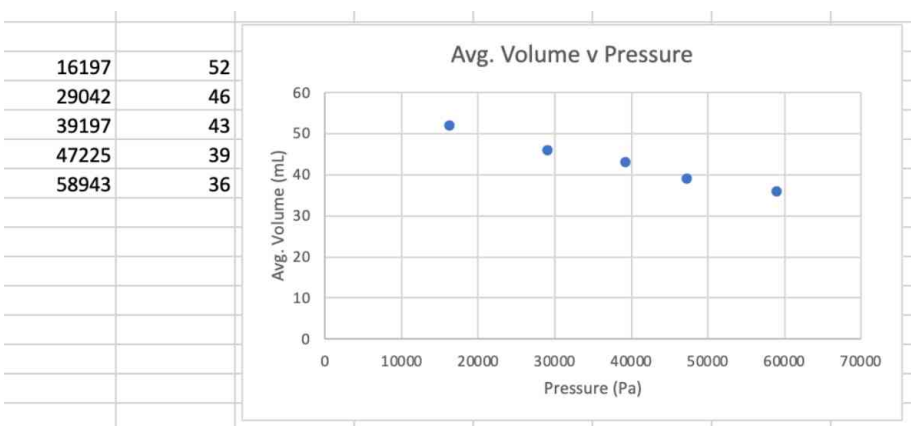
Book #	Volume: Trial 3 (mL)
1	53.mL
1,2	45.mL
1,2,3	42.mL
1,2,3,4	39.mL
1,2,3,4,5	36.mL

Book #	Average Volume V_{avg} (mL)
1	52.mL
1,2	46.mL
1,2,3	43.mL
1,2,3,4	39.mL

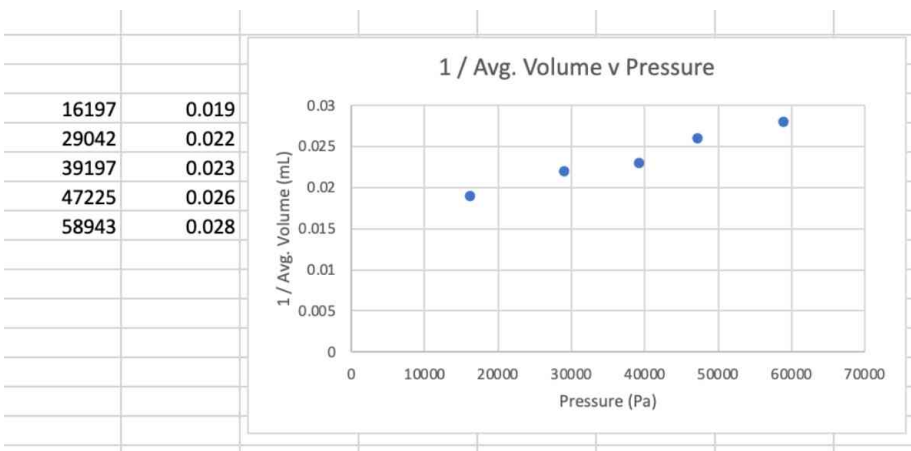
1,2,3,4,5	36.mL
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Book #	1/Average Volume $1/V_{\text{avg}}$ (1/mL)
1	0.019
1,2	0.022
1,2,3	0.023
1,2,3,4	0.026
1,2,3,4,5	0.028

Graph 1: Average Volume vs. Pressure



Graph 2: 1/Average Volume vs. Pressure



Competency Review

Gas particles in a closed system ____.

- ☐ are in constant motion
- ☐ move randomly
- ☐ collide with one another and the sides of the container
- ☒ All of the above

A graph that describes the relationship of volume versus the inverse value of pressure ($1/P$) displays a linear relationship.

- ☒ True
- ☐ False

The International System unit of force is the ____, and the unit of pressure is the ____.

- ☒ Newton; Pascal
- ☐ Pascal; Newton

A syringe is filled with gas and sealed to create a closed system. What will happen when the plunger of the syringe is depressed?

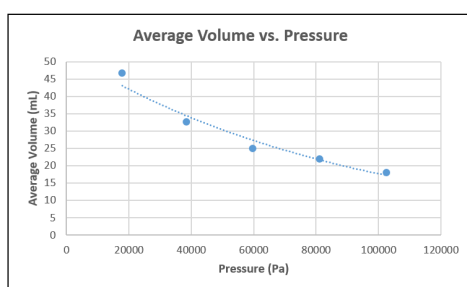
- ☐ The volume of the gas will increase.
- ☐ The number of gas particles will decrease.
- ☐ Gas particles will have an increased rate of collisions.
- ☒ Pressure will decrease slightly.

The diameter and radius of the syringe were measured to determine the ____ of the Boyle's law apparatus.

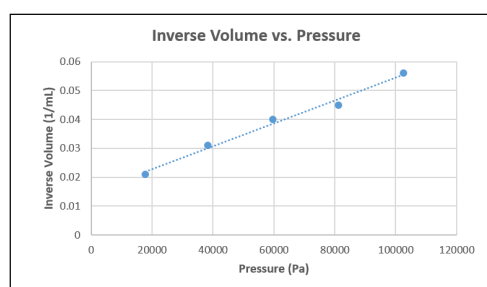
- ☒ surface area
- ☐ volume
- ☐ mass
- ☐ All of the above

Which scatter plot describes a linear relationship?

A:



B:



- ☐ A
- ☐ B
- ☒ Both A and B
- ☐ Neither A nor B

At a constant temperature the volume of a gas is increased ten times, resulting in the pressure of the gas ____.

- ☒ decreasing by 10 times
- ☐ decreasing by 20 times
- ☐ increasing by 10 times
- ☐ remaining constant

Extension Questions

The Ideal Gas Law has more variables than Boyle's Law, and it contains the ideal gas constant, R . The ideal gas law is $PV = nRT$, where T is the absolute temperature and n is the number of moles of gas present in a system.

- a. In Boyle's Law, $PV = k$ (a constant). Rearrange the Ideal Gas Law so that the constant R is similarly equal to the variables.
- b. If $PV = k$, and $PV = nRT$ as well, express k in terms used in the Ideal Gas Law.
- c. If you took laboratory syringe setup outside in the summer or put it in the refrigerator, would Boyle's Law be able to adequately describe what would happen to the volume of the system? Why?
- d. There are other gas laws aside from Boyle's Law, including: Charles' Law ($V/T = k$), Gay-Lussac's Law ($P/T = k$), and Avogadro's Law ($V/n = k$). All these laws feature a " k " value (a constant). What is different about k in these laws compared to the constant R in the Ideal Gas Law?

- (a) $R = PV/nT$
(b) $k = nRT$
(c) I'm going to say no, it cannot adequately describe what would happen to the volume because we would need something to push down on the top of the syringe setup. I don't think a change in environment is enough to create a force like that. Then again, maybe there are some environments that can!
(d) The constant k in these equations is a constant based off of conditions from other variables to where the constant R is a constant with an assigned value that doesn't change. I hope I didn't just confuse things more..